Reading Activity 3: Feature Selection for Regression Problems

1. Introduction

The objective of this activity was to analyze a dataset of secondary school student performance in mathematics (student-mat.csv) and identify the most significant features contributing to the final grade (G3). Using Python, I performed exploratory data analysis, applied Lasso Regression for feature selection, and compared the performance of a complex regression model against a simple baseline model.

2. Exploratory Data Analysis (EDA)

2.1 Distribution of Target Variable (G3)

The target variable for this regression task is G3 (Final Grade, range 0-20). My analysis of the distribution reveals that grades are roughly normally distributed around a mean of 10.4. However, there is a notable spike at 0, representing students who likely dropped out or missed the final exam.

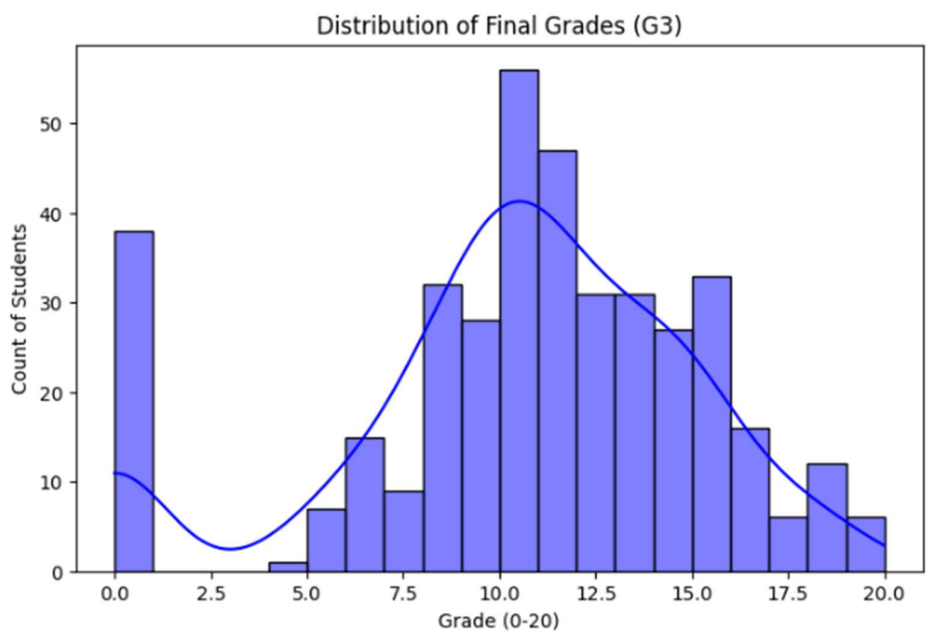

Figure 1: Distribution of Final Grades (G3).

2.2 Correlation Analysis

To identify potential predictors, I analyzed the correlation of all features with G3. The analysis highlights that a student's history is the strongest predictor of their future success.

- **Strongest Positives:** G2 (Second Period Grade) and G1 (First Period Grade) have extremely high correlations (> 0.80), followed by Medu (Mother's Education).
- **Strongest Negatives:** failures (Past Class Failures) and age showed the strongest negative correlations.

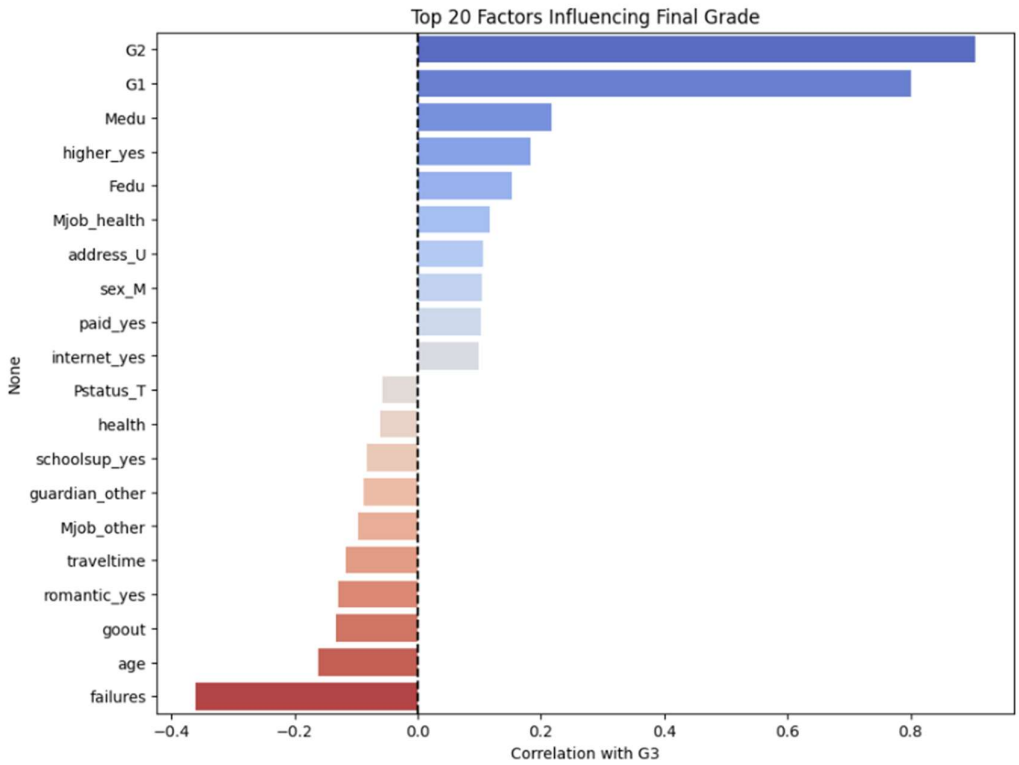

Figure 2: Correlation of features with Final Grade.

3. Feature Selection Analysis

To select the most relevant features and avoid overfitting, I employed Lasso Regression (L1 Regularization). Lasso is a technique that shrinks the coefficients of less important features to exactly zero, effectively removing them from the model.

- Input: All 30+ original variables (scaled using StandardScaler).
- Result: Lasso reduced the dataset to 18 significant features, eliminating noise variables such as nursery and paid classes.

The most critical features identified by the Lasso model were:

- **G2 & G1** (Dominant impact)
- **Absences** (Surprisingly selected as a top feature)

- **Failures** (Negative impact)

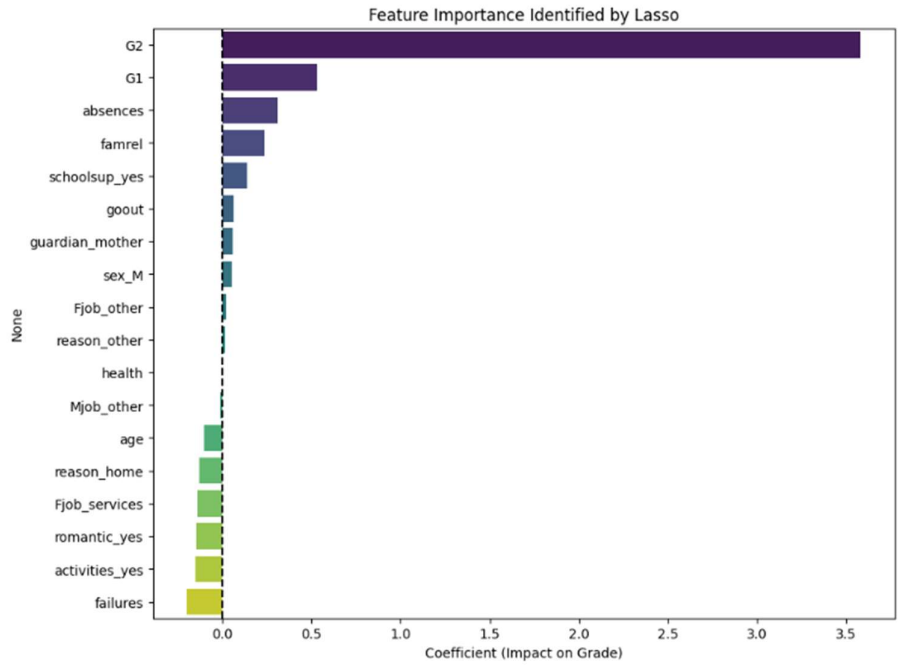

Figure 3: Lasso Regression Feature Importance.

4. Model Evaluation & Results

I trained two Linear Regression models to test if the features selected by Lasso could outperform a simple baseline.

- Model A (Complex): Trained using the 18 features selected by Lasso.
- Model B (Baseline): Trained using only G1 and G2 (Past Grades).

Performance Comparison Table

Model	RMSE (Lower is Better)	R^2 Score (Higher is Better)
Model A (Selected Features)	2.3365	0.7338
Model B (G1 & G2 Only)	2.0523	0.7946

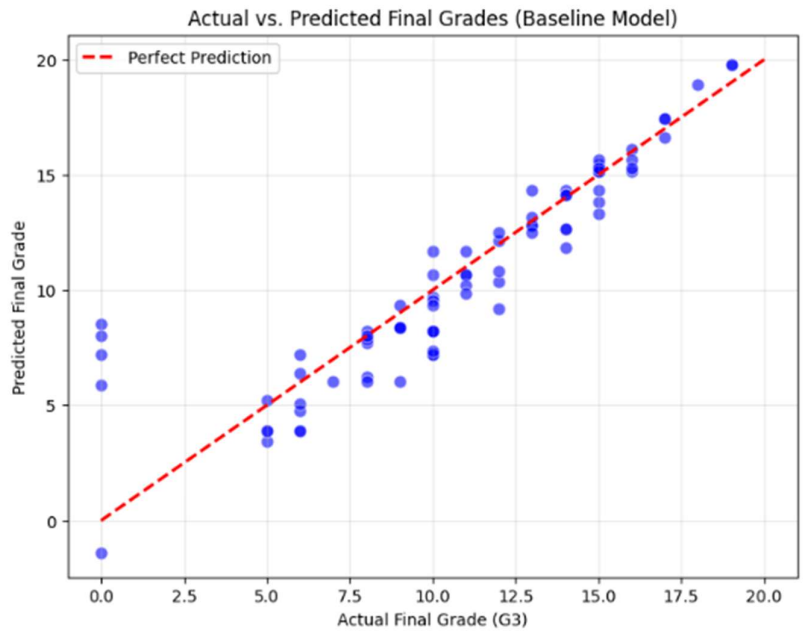

Figure 4: Actual vs. Predicted Grades (Baseline Model).

5. Conclusion

My analysis yielded a counter-intuitive but insightful result: The simple baseline model outperformed the complex model.

While Lasso successfully identified statistically significant socio-economic and behavioral factors (like Medu, goout, and absences), adding these features introduced "noise" that slightly degraded the prediction accuracy compared to using past grades alone.

In this specific dataset, a student's past academic performance (G1 and G2) is such a dominant predictor that it effectively captures the influence of all other variables. For predictive purposes, knowing a student's previous grades is sufficient; however, for intervention purposes, factors like failures and Medu remain important areas of focus.